

Name: AVIRAJ KUMAR RAI AVIRAJ KUMAR RAI | Reg. No: RA2511003011026

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## Unit – 1 : Session – 7 : SLO – 2 | Class Design & Inheritance

### ■ Activity (Solutions)

- 1) class Vehicle { String brand; void move(){ System.out.println("moving"); } } class Car extends Vehicle { int doors; } class ElectricCar extends Car { int batteryKwh; void charge(){ System.out.println("charging"); } } — multilevel hierarchy Vehicle → Car → ElectricCar.
- 2) class Employee { String name; double salary; void work(){ System.out.println("working"); } } class Manager extends Employee { String department; void conductMeeting(){ System.out.println("meeting"); } } Manager m = new Manager(); m.work(); m.conductMeeting();
- 3) Overriding + super: class Employee { void show(){ System.out.println("Employee"); } } class Manager extends Employee { @Override void show(){ super.show(); System.out.println("Manager"); } } new Manager().show(); prints Employee then Manager — super.show() invokes the superclass version, letting the subclass extend rather than fully replace the parent behavior.